IMAGE INCLUSION AFTER APPLYING THE FORGETFUL FUNCTOR LIFTS TO A FACTORIZATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a type of algebraic structure (C, F): F preserves fibre products and reflects isomorphisms. invoked at proof lines 1493-1496. Provides the identification F(A times_B C) = F(A) and the conclusion A = A times_B C.

Lemma 0.2. *Let $(\mathcal{C}, F)$ be a type of algebraic structure. Suppose that $A, B, C \in \mathrm{Ob}(\mathcal{C})$. Let $f : A \to B$ and $g : C \to B$ be morphisms of $\mathcal{C}$. If $F(g)$ is injective, and $\mathrm{Im}(F(f)) \subset \mathrm{Im}(F(g))$, then f factors as $f = g \circ t$ for some morphism $t : A \to C$.*

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Date: stacks-project:sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.